Data and ETL scripts (outside the runner)

MLHD+ listening histories, the MusicBrainz database and the Spotify catalogue, converted to Parquet and bridged into one item space.

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Preparation

P1	corpus audit	**P2**	train, validation, test split	**P2.5**	training table, statistics
P3	sampling tables				

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Features and baselines

P4	item feature matrices	**P5**	evaluation harness, popularity	**P6**	ALS baseline
P7	item-item relational graph	**P8**	extra fusion features	**P9**	item co-occurrence
P9b	exact co-listening neighbours	**P10**	user profile vectors	**ALS-D**	ALS phase D tuning
KNN	ItemKNN baseline				

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Training pairs and model fits

P11	training pairs	**P12**	concept features of the pairs	**P13a**	audit and base GAM fit
P13b	experiment arms	**P11b**	more negatives	**P12b**	their concept features
P13b2	loss form, negatives	**Diag**	GAM against ItemKNN slices	**Refit**	GAM refit, 2 × 2 design

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Stability and the frozen model

P13c	bootstrap stability	**P13d**	bounded content shapes	**P13c**	stability of the final model

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Evaluation

P14	cold-track GAM	**P14**	evaluation on the eval listeners	**Curve**	cold-start curve
Beyond	novelty, coverage, diversity	**Removal**	evidence removal of the reasons		

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Plain scripts and the figures notebook, on the stored outputs

lb_demo.py	ListenBrainz demo	**lb_outside.py**	tracks outside MLHD	**tree_audit.py**	tree-GAM audit
figures.py	thesis tables	**notebook**	thesis figures		

Grey: no evaluation stage depends on these stages (not used by the final model). Arrows: the order of the phases.